Linguistic Convergence in Guardian Business Journalism Following LLM Public Release: A Multilevel Analysis, 2019–2025

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Abstract

This study investigates between-author linguistic convergence in Guardian Business journalism. The period examined coincides with mass public LLM adoption, operationalised as the release of ChatGPT in November 2022. Using a corpus of 14847 articles from 45 authors spanning 2019 to 2025, we employ a multilevel modelling framework with random intercept and random slope specifications to examine linguistic convergence across seven features: average sentence length, burstiness, hedge rate, moving average type-token ratio and contrastive, elaborative and inferential discourse markers. Between-author variance decomposition provides partial evidence of linguistic change coinciding with the post-ChatGPT period, with hedge rate showing a consistent convergence pattern across both corpus-level and individual trajectory analysis. Random slope models indicate that authors respond heterogeneously to the post-ChatGPT period. These findings provide partial evidence of linguistic convergence, but this is gradual and feature-specific rather than uniform.

Keywords: linguistic, convergence, between-author variance, heterogeneous, discourse markers, hedging, multilevel modelling, large language models, ChatGPT, Guardian Business journalism

1. Introduction

Since the public release of ChatGPT 3.5 in November 2022, concerns emerge that large language models influence the style of professional writing. LLMs generate text by predicting statistically probable continuations of their training data, optimised through reinforcement learning from human feedback (RLHF) to produce fluent, broadly acceptable prose. This process inherently draws output toward a linguistic mean. As journalists become increasingly exposed to or reliant on such output, the distinctiveness of individual authorial voices is at risk. Journalists therefore represent a particularly consequential test case.

Sourati et al. (2026, p.2) synthesise evidence across linguistics, psychology, and cognitive science, arguing that as LLMs become embedded in daily life, they risk standardising language and reasoning, marginalising alternative voices, and flattening the cognitive landscapes that drive collective intelligence. However, their theoretical argument lacks empirical verification. Ansari et al. (2026, p.1) identify a substantial increase in LLM-generated content across journalism, including The Guardian, with AI-associated writing exhibiting more uniform linguistic patterns than human-authored text. However, their approach detects whether LLMs are being used and does not address whether that use, or exposure to LLM-influenced writing, produces convergence between authors over time. Filimonovic et al. (2026, p.6) identifies statistically significant linguistic convergence in AI-assisted scientific publications toward U.S. scientific English post-ChatGPT. In this case however, convergence is measured against an external linguistic benchmark rather than

between writers themselves. This study addresses these gaps by investigating whether linguistic convergence is detectable between journalists in the same domain (Guardian Business) following the public release of ChatGPT.

Two mechanisms may explain how LLM exposure associates with linguistic convergence. First, when journalists use LLMs to draft or edit their writing, their output may shift toward LLM linguistic defaults. Second, even without direct use, repeated passive exposure to LLM-generated content may gradually shape writing habits over time. Regardless of which mechanism is operating, Rogers' (1962, p.23) diffusion of innovation framework suggests that technological adoption follows a sequential pattern across a population rather than occurring simultaneously. Applied to our study, given that our observation window spans only three years post-release, if journalists are adopting LLM-assisted writing at different rates, we predict we may observe only early-stage heterogeneous and bidirectional author trajectories rather than uniform convergence. This study does not seek to attribute convergence to either mechanism directly, but tests whether a converging signal is detectable in Guardian Business journalism following the mass public release of LLMs.

We thus propose two hypotheses. First, we predict evidence of reduced between-author variance, specifically in surface-level syntactic and epistemic features, consistent with the RLHF process described above. Second, consistent with the diffusion of innovation framework, we predict that author responses will be heterogeneous rather than uniform.

We address two questions:

1. Is there evidence of reduced between-author variance in Guardian Business journalism during the period coinciding with the public release of ChatGPT in November 2022?
2. Do authors shift uniformly or heterogeneously across linguistic features in the post-release period?

This study targets a multidisciplinary readership spanning quantitative linguistics, journalism practice, and AI governance, each of whom may find the empirical framework and findings relevant to their respective concerns.

This paper is organised as follows. Section 2 describes the data, variables, and model specifications. Section 3 presents results for each research question. Section 4 discusses findings, limitations, future research and implications. Section 5 is the conclusion.

2. Methodology

2.1 Data

The dataset used in this study is constructed from articles published by *The Guardian*, accessed via the official Guardian Open Platform API[1]. The Guardian is a major UK news outlet, and restricting the dataset to a single publication helps control for variation in editorial style across sources.

To maintain consistency in tone and subject matter, the analysis is restricted to the *Business* section. The *Business* section was chosen over the Opinion section for three reasons. First, business journalism operates within a consistent editorial register: reporters write to shared factual standards, whereas opinion pieces vary widely in structure, from bullet points to polemical prose, making cross-author comparison unreliable. Second, business journalism involves processing large amounts of financial data, earnings reports, and economic statistics into readable prose. This is exactly the kind of repetitive, information-dense task LLMs are used for. Third, all authors write within the same broad domain, ensuring that any linguistic variation reflects author-level differences rather than topic-driven genre shifts.

Articles are retrieved with the following constraints:

1. Section: *Business*
2. Time period: January 2019 – December 2025
3. Fields extracted: headline, author (byline), and full article text (bodyText)
4. Tags: type, keyword, and tone metadata

The choice of time frame spans multiple years before and after November 2022, enabling a comparison of baseline linguistic behaviour and potential changes following the release of ChatGPT.

To ensure the dataset consists of comparable units of writing, we apply the following series of preprocessing and filtering steps:

1. Article type filtering:
   These filters restrict the dataset to standard news reporting by excluding formats with distinct structural and linguistic conventions.
   - Retain only standard news articles (‘type/article’ in tags).
   - Exclude live blogs (‘tone/minutebyminute’ in tags).
   - Exclude opinion-based content (‘tone/comment’ in tags).

[1] The Guardian, Content API Documentation, https://open-platform.theguardian.com/documentation/

2. Author filtering:

   These filters ensure consistent author identification for author-level analysis by removing missing values and institutional authors, standardising author names, and retaining only sufficiently represented authors with observations in both periods.

   - Remove entries with missing authors.
   - Retain only authors contributing at least 1% of the total sample.
   - Remove role descriptors and location-based suffixes (e.g. "Economics correspondent", "in New York").
   - Exclude institutional authors (e.g. "Guardian staff").
   - Retain only authors with articles in both pre- and post-November 2022 periods.

After cleaning, the dataset consists of 14,847 articles written by 45 journalists over the 2019 to 2025 period.

2.2 Variables

Let y_{ij} denote the standardised value of a linguistic outcome for an article i written by author j. All variables are defined at the article level unless otherwise stated.

Outcome variables:

1. Average Sentence Length (ASL_{ij})

   Let article i written by author j contain S_{ij} sentences, indexed with $s = 1,.., S_{ij}$ with sentence lengths l_{ijs}. Then,

$$ASL_{ij} = \frac{1}{S_{ij}} \sum_{s=1}^{S_{ij}} l_{ijs}$$

   This computes the mean sentence length by averaging the number of tokens across all sentences in the article, measuring the average number of tokens per sentence.

2. Burstiness ($Burst_{ij}$)

$$Burst_{ij} = \frac{\sigma(l_{ijs})}{\mu(l_{ijs})}$$

   This computes the coefficient of variation of sentence lengths by dividing the standard deviation by the mean, measuring variability in sentence structure within an article.

3. Moving Average Type-Token Ratio ($MATTR_k_{ij}$):

Let N_{ij} denote the number of tokens in article i, and let w_{ij1}, ..., $w_{ijN_{ij}}$ denote the sequence of tokens. Then, for a window size k:

$$MATTR_k_{ij} = \frac{1}{N_{ij}-k+1} \sum_{t=1}^{N_{ij}-k+1} \frac{|\{w_{ijt}, \ldots, w_{ij,t+k-1}\}|}{k}$$

This computes the average proportion of unique tokens within rolling windows of length k across the article, measuring lexical diversity while reducing sensitivity to document length.

4. Discourse Marker Rates ($DMRate_{ijc}$)

   For category $c \in$ {contrastive, elaborative, inferential}, let DM_{ijc} be the total discourse markers of type c in article i. Then,

$$DMRate_{ij} = \frac{DM_{ijc}}{N_{ij}} \times 1000$$

   This computes the frequency of discourse markers by dividing their count by total tokens and scaling per 1,000 tokens, measuring the use of rhetorical connectors within the text.

5. Hedge Rates ($Hedge\ Rate_{ij}$)

   Let H_{ij} be the total hedge expressions in article i. Then,

$$Hedge\ Rate_{ij} = \frac{H_{ij}}{N_{ij}} \times 1000$$

   This computes the frequency of hedge expressions by dividing their count by total tokens and scaling per 1,000 tokens, measuring the degree of epistemic uncertainty in the text.

Predictors:

1. Post-ChatGPT release indicator ($post_{ij}$):

   A binary variable equal to 1 if article i is published from November 2022 onwards, and 0 otherwise.

2. Article length ($log_word_count_{ij}$, $log_word_count_{j}$):

   The natural logarithm of total token count, decomposed into a within-author component (each article's log word count demeaned by that author's mean) and a between-author component (each author's mean log word count). Both components are included in the fixed effects model to partial out compositional differences in article length that could otherwise confound the post-release estimate as longer articles mechanically produce higher feature counts regardless of authorial style.

2.3 Operationalisation of linguistic features

This study processes all articles using the spaCy natural language processing library. Sentence boundaries and tokens are identified with spaCy's parser.

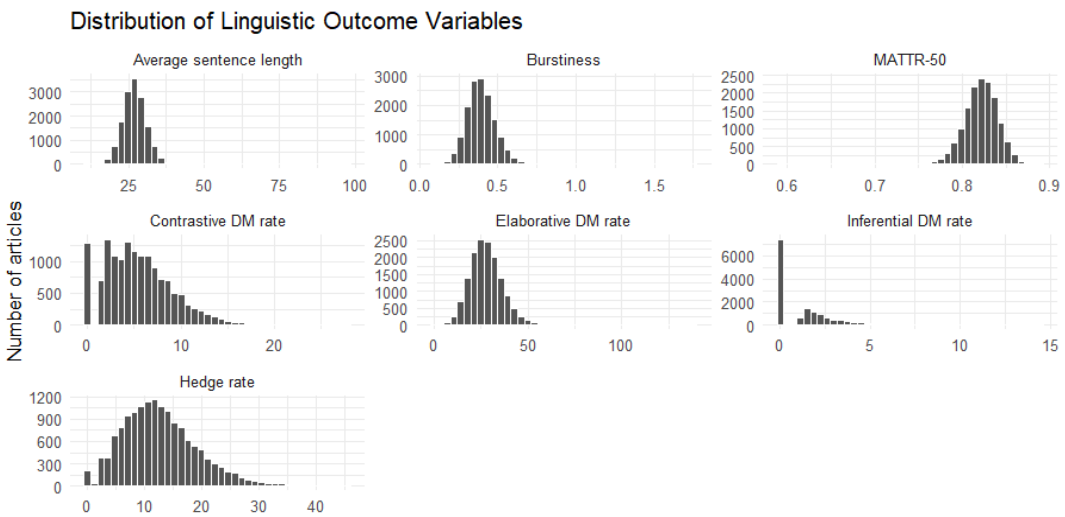
This study identifies discourse markers using a dictionary-based approach following Fraser (2009, p.301), grouped into contrastive, elaborative, and inferential categories (word list provided in Appendix A). Phrase matching is applied to detect both single-word and multi-word expressions, with matches identified based on case-insensitive token sequences. This study identifies hedge expressions (word list provided in Appendix B) using a lemma-matched dictionary constructed according to the concept of epistemic metadiscourse proposed by Hyland (2010, p.129), including modal verbs, adverbs, attribution verbs, and adjectival forms. Lemma matching ensures consistent treatment of different inflected forms of the same word.

The construction of dictionary-based measures may be sensitive to the selection of lexical items within each category. Furthermore, the MATTR measure of lexical diversity depends on the choice of window size. This study adopts a window size of 50 tokens, and alternative specifications may yield different estimates.

2.4 Descriptive Statistics

Figure 1 presents the distributions of all seven outcome variables. Most variables display moderate right-skewness, which is typical of rate-based linguistic measures; at the corpus size of 14,847 articles, this does not pose material concerns for the multilevel framework. Inferential DM rate is an exception, with approximately 50% of articles recording zero values and a skewness of 1.69, warranting the log(x+1) transformation described in Section 2.5. Full summary statistics are provided in Table A1, Appendix C.

Figure 1: Distribution of Linguistic Outcome Variables



2.5 Data Transformation

Prior to standardisation, inferential DM rate is log-transformed to address zero-inflation, as approximately 50% of articles record zero inferential markers. The transformation applied is:

$$\widetilde{X}_{ij} = log(X_{ij} + 1)$$

where the constant 1 is added to retain zero-valued observations. All outcome variables, including the log-transformed inferential DM rate, are then standardised prior to estimation:

$$y_{ij} = \frac{X_{ij} - \overline{X}}{\sigma_X}$$

where X_{ij} denotes the raw feature and $\overline{X}$, σ_X are the sample mean and standard deviation.

2.6 Multilevel Model Justification

The data structure of this study is hierarchical, comprising 14,847 articles nested within 45 authors, each observed over a six-year period from 2019 to 2025. The most comparable convergence study, Filimonovich et al. (2026), employs OLS regression with fixed effects. However, this investigation of between-author variance requires a different approach. Multiple articles by the same author are not independent observations, and treating them as independent conflates within-author variation with between-author variation, leading to unreliable standard errors.

Multilevel modelling addresses this by partitioning variance into within-author variation across articles and between-author variation across the corpus. This decomposition is essential to the research design, as the primary outcome of interest is the between-author variance and whether it is statistically significantly lower in the post-ChatGPT period. This study estimates all models using the lme4 and lmerTest packages in R, with REML set to FALSE to enable likelihood ratio tests for model comparison across nested model specifications.

2.7 Model Specifications

2.7.1 Model 1: Null Model

The null model is specified as a two-level random effects variance components model, where articles i are nested within authors j.

$$y_{ij} = \beta_0 + u_j + e_{ij} \tag{1}$$

where y_{ij} denotes the standardised outcome variable, β_0 is the overall mean across all authors of the outcome variable, u_j is the author-level random intercept, assumed $u_j \sim N(0, \sigma_u^2)$, and e_{ij} is the article-level residual, assumed $e_{ij} \sim N(0, \sigma_e^2)$. σ_u^2 is the between-author variance and σ_e^2 is the within-author variance.

The intraclass correlation coefficient is defined as:

$$ICC = \frac{\hat{\sigma}_u^2}{\hat{\sigma}_u^2 + \hat{\sigma}_e^2} \tag{2}$$

which quantifies the proportion of total variance attributable to author-level differences. An ICC above 5% indicates meaningful author-level clustering, justifying multilevel modelling.

2.7.2 Model 2: Fixed Effects Model

The model aims to investigate whether linguistic features shift coincides with the post-release period. It extends the specification in the null model, including article-level covariates:

$$y_{ij} = \beta_0 + \beta_1 post_{ij} + \beta_2^W log_word_count_{ij} + \beta_2^C log_word_count_j + u_{0j} + e_{ij} \tag{3}$$

where y_{ij}, β_0, u_j, e_{ij} are interpreted the same as equation (1). β_1 is the fixed effect of the post-release period, representing the estimated mean shift in each linguistic feature post-ChatGPT, after accounting for article length. β_2^W and β_2^C capture the within- and between-author effects of article length respectively, as defined in Section 2.2.

To assess the extent to which the fixed effects explain variation in outcome variables, we compute the proportional differences in variance relative to the null model:

$$R_1^2 = \frac{\hat{\sigma}_{e(M1)}^2 - \hat{\sigma}_{e(M2)}^2}{\hat{\sigma}_{e(M1)}^2}, R_2^2 = \frac{\hat{\sigma}_{u(M1)}^2 - \hat{\sigma}_{u(M2)}^2}{\hat{\sigma}_{u(M1)}^2} \tag{4}$$

R_1^2 measures the proportion of within-author variance explained and R_2^2 measures the proportion of between-author variance explained. While R_1^2 captures within-author variation attributable to article-level covariates, this study is primarily concerned with R_2^2. A positive R_2^2 indicates that between-author variance is smaller in Model 2 than in the null, meaning authors are more similar to

one another when the post-release period and article length are jointly accounted for. This is interpreted as consistent with linguistic convergence, though it cannot be attributed to the post-release period alone. . A negative R_2^2 indicates divergence. Model fit is assessed through a likelihood ratio test comparing Model 2 against Model 1, under H_0: $\beta_1 = \beta_2^W = \beta_2^C = 0$. The test statistic follows a $\chi^2(3)$ distribution and H_0 is rejected if $p < 0.05$.

2.7.3 Model 3: Random Slope Model

Model 2 serves as the primary basis for addressing RQ1, with R_2^2 quantifying the reduction in between-author variance attributable to the post-release period. However, R_2^2 assumes a uniform author response, it cannot capture whether individual authors responded differently. Model 3 relaxes this assumption by introducing a random slope on $post$, allowing each author's response to deviate from the corpus average.

$$y_{ij} = \beta_0 + \beta_1 post_{ij} + \beta_2^W log_word_count_{ij} + \beta_2^C log_word_count_j + u_{0j} + u_{1j} post_{ij} + e_{ij} \quad (5)$$

where y_{ij}, β_0, β_1, β_2^W, β_2^C, e_{ij} are interpreted the same as equation (3). $u_{0j} \sim N(0, \sigma_{u0}^2)$ is the author-level random intercept, where σ_{u0}^2 is the between-author variance in baseline linguistic features. $u_{1j} \sim N(0, \sigma_{u1}^2)$ is the random slope for author j, capturing author-specific deviation from the average pre-to-post shift, where σ_{u1}^2 is the between-author variance in the magnitude of that shift.

To quantify the spread of individual author-level post-release slopes, the plausible values range (PVR) is computed as:

$$PVR = \hat{\beta}_1 \pm 1.96\sqrt{\hat{\sigma}_{u1}^2} \quad (6)$$

providing the interval within which 95% of authors' individual post-release slopes are expected to fall. Model fit is assessed through a likelihood ratio test comparing Stage 3 model against Stage 2, under $H_0 = \sigma_{u1}^2$. The test statistic follows a $\chi^2(2)$ distribution and H_0 is rejected if $p < 0.05$.

2.8. Model Assumptions

Normality of residuals and random effects was assessed through QQ plots across both Models 2 and 3. Random effects, including random slopes, displayed approximate normality across all outcomes. Residuals showed mild to moderate right-skewness consistent with the distributions described in Section 2.4, which at this sample size does not pose concerns for inference. The inferential DM rate residuals showed residual zero-inflation following log transformation, elaborative DM rate residuals exhibited a cluster of upper-tail outliers, and MATTR-50 residuals exhibited a small number of extreme lower-tail outliers, all of which are acknowledged as limitations of the current operationalisation.

3. Results

3.1 RQ1: Is there evidence of reduced between-author variance in Guardian Business journalism during the period coinciding with the public release of ChatGPT in November 2022?

Null models are estimated for each of the seven outcome variables (equation 1), with ICCs computed using equation (2) and reported in Table 1. All outcomes return ICC values substantially above 5%, confirming meaningful author-level clustering (range: 7.2% - 27.2%), thus justifying the multilevel modelling approach. Hedge rate shows the smallest between-author variation (7.2%), suggesting hedging behaviour varies more within authors' own writing than between them. To further characterise this clustering, caterpillar plots of the estimated random intercepts are provided in Appendix D , with Figure 2 presenting average sentence length as a representative example. The widespread separation of confidence intervals confirms statistically significant author-level differences across the corpus.

Fixed effects models are estimated for each outcome (equation 3), with results reported in Table 2. For all outcomes except elaborative DM rate, the fixed-effects model significantly improves fit over the null (LRT, $p < 0.001$), indicating that the post-release period and article length jointly explain linguistic variation. R_1^2 values are small across all outcomes (range: 0.000–0.084), indicating that within-author variation is largely not explained by the fixed effects.

Three outcomes show statistically significant post-release associations. Average sentence length increased post-release by 0.062 standard deviations ($\hat{\beta}_1 = 0.062,\ p < 0.001$), and hedge rate decreased by 0.063 standard deviations ($\hat{\beta}_1 = -0.063,\ p < 0.001$), suggesting a shift toward more assertive language. For both outcomes, R_2^2 is positive (0.119 and 0.075 respectively), indicating a

reduction in between-author variance when the post-release period and article length are jointly accounted for. As R_2^2 reflects the joint contribution of all predictors, the significant $\hat{\beta}_1$ indicate that a post-release shift is present even after accounting for article length differences. MATTR-50 also increased by 0.034 standard deviations ($\hat{\beta}_1 = 0.034$, $p = 0.040$), though $R_2^2 = 0.001$ indicates virtually no reduction in between-author variance, suggesting a uniform mean shift without convergence; possible explanations are discussed in Section 4. The remaining four outcomes show no statistically significant post-release association. Their R_2^2 values, while positive for some, are better accounted for by article length, as evidenced by the significant article length coefficients reported in Appendix E. These outcomes are therefore not interpreted further in terms of convergence.

Together, these results provide partial evidence of reduced between-author variance coinciding with ChatGPT's release. Of the seven outcomes, only average sentence length and hedge rate show both a significant post-release shift and a positive R_2^2, consistent with modest linguistic convergence in these two features. The remaining outcomes show no such pattern, suggesting that any homogenisation effect is feature-specific rather than corpus-wide.

Figure 2: Author-level Random Intercepts for Average Sentence Length, Null Model

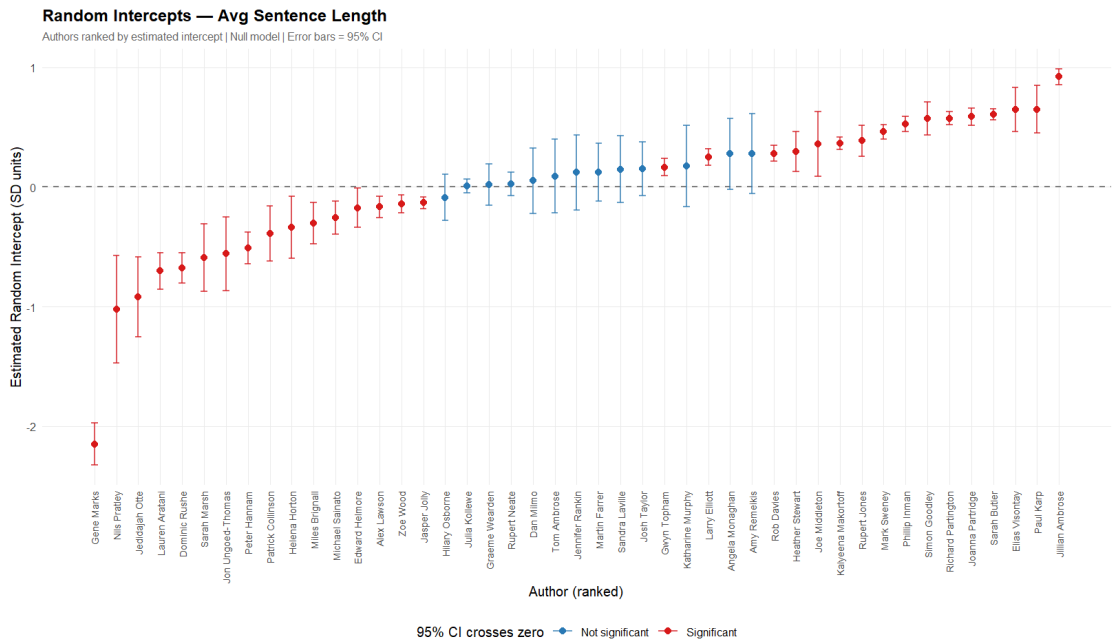


Table 1: Null Model Variance Components and Intraclass Correlation Coefficients by Outcome Variable

Outcome	**ICC**
Average Sentence Length	0.272
Burstiness	0.234
Contrastive DM rate	0.134
Elaborative DM rate	0.113
Inferential DM rate	0.113
Hedge rate	0.072
MATTR-50	0.187

Note: ICC values are expressed as proportions; in results we multiply by 100 for the percentage of total variance attributable to between-author differences.

Table 2: Stage 2 Fixed Effects Model Results

Outcome	**Estimated Coefficient** $post$ $(\hat{\beta}_1)$	**Standard Error**	**p-value** $post$	R_1^2	R_2^2	**LRT vs Null (p-value)**
Average Sentence Length	0.062	0.016	< 0.001	0.028	0.119	< 0.001
Burstiness	0.003	0.016	0.863	0.084	0.225	< 0.001
Contrastive DM rate	0.023	0.017	0.172	0.030	0.096	< 0.001
Elaborative DM rate	-0.003	0.017	0.838	0.000	0.003	0.943
Inferential DM rate	-0.007	0.017	0.659	0.045	0.403	< 0.001
Hedge rate	-0.063	0.017	<0.001	0.005	0.075	< 0.001
MATTR-50	0.034	0.016	0.040	0.003	0.001	< 0.001

Note: $\hat{\beta}_1$ *is the estimated effect of the post-release period indicator after controlling for article word count and author-level baseline differences.*

3.2 RQ2: Do authors shift uniformly or heterogeneously across linguistic features in the post-release period?

Random slope models are estimated for each of the seven outcome variables (equation 5), with results reported in Table 3. Likelihood ratio tests were significant for six out of seven outcomes ($p < 0.05$), with inferential DM rate being the exception ($p=0.152$). Notably, *burstiness, contrastive,* and *elaborative discourse marker rates*, which showed no significant mean-level shift in Model 2, all exhibit significant random slopes, suggesting the corpus average was masking considerable variation at the author level.

The between-author slope variance $\hat{\sigma}^2_{u1}$ is largest for average sentence length (0.041) and hedge rate (0.034), indicating the greatest heterogeneity in author-level responses. The plausible values range crosses zero for all six significant outcomes, confirming that no linguistic feature exhibits a uniformly directional shift and for every feature, some authors shifted positively while others shifted negatively. The widest ranges are observed for hedge rate [−0.423, +0.298] and average sentence length [−0.336, +0.460], reflecting the greatest spread in individual author trajectories. The narrowest range is burstiness [-0.189, +0.194], suggesting more modest but still bidirectional heterogeneity.

The correlation between each author's random intercept and random slope, cor(u_0, u_1), captures whether authors with higher baseline levels tended to increase or decrease post-release. A negative correlation indicates convergence: authors who were high on a feature pre-release tended to decrease, while those who were low tended to increase, narrowing the gap between them. A positive correlation indicates divergence: high-baseline authors increased further while low-baseline authors decreased further, widening between-author differences. Burstiness (−0.326), hedge rate (−0.261), and MATTR-50 (−0.340) display negative correlations, suggesting convergence. In contrast, average sentence length (0.231), elaborative DM rate (0.162), and contrastive DM rate (0.639) exhibit positive correlations, indicating divergence.

These results indicate heterogeneous author-level responses across all seven linguistic features in the period coinciding with ChatGPT's release. The implications for the convergence hypothesis are addressed in the discussion.

Table 3: Random Slope Model Results

Outcome	Between-author slope variance $\hat{\sigma}^2_{u1}$	Correlation of intercept and slope $cor(u_{0j}, u_{1j})$	Plausible Values Range	LRT vs Null (p-value)
Average Sentence Length	0.041	0.149	[-0.336, +0.460]	<0.001
Burstiness	0.010	-0.326	[-0.189, +0.194]	<0.001
Contrastive DM rate	0.032	0.636	[-0.326, +0.371]	<0.001
Elaborative DM rate	0.014	0.162	[-0.234, +0.227]	0.002
Inferential DM rate	0.004	-0.0071	[-0.129, +0.114]	0.152
Hedge rate	0.034	-0.260	[-0.423, +0.298]	<0.001
MATTR-50	0.017	-0.336	[-0.220, +0.288]	<0.001

Note: $cor(u_{0j}, u_{1j})$ is the estimated correlation between-author baseline and their post-release slope; negative values indicate a convergence pattern, positive values indicate divergence.

4. Discussion

4.1 Interpretation of findings

This study investigates whether between-author variance in linguistic features among Guardian Business journalists reduced in the period coinciding with ChatGPT's public release, and whether author responses to this period were heterogeneous. The results provide partial and mixed evidence of linguistic convergence. Across authors on average, sentence length and hedge rate show both a significant mean-level change and a reduction in between-author variance in the post-release period, when article length is jointly considered. However, at authors' individual trajectories, these two outcomes tell different stories. Hedge rate shows a convergence pattern in author trajectories, while average sentence length shows divergence, with authors who already wrote longer sentences tended to increase their length further post-release, while shorter-sentence authors moved in the opposite direction. Across all seven features, author responses are heterogeneous, with no linguistic feature exhibiting uniform directional shift. Together, these findings suggest a selective and uneven process of linguistic change rather than full-scale convergence.

Our first hypothesis predicted reduced between-author variance in surface-level syntactic and epistemic features. This prediction is partially confirmed. For average sentence length and hedge rate, the post-release period is associated with both a significant mean-level change and reductions in between-author variance of 11.9% and 7.5% respectively, when article length is jointly considered. No such combination is observed for burstiness, contrastive, and inferential DM rates, whose variance reductions are better explained by article length. Elaborative DM rate shows neither a significant mean-level shift nor a variance reduction. MATTR-50 presents a distinct pattern, showing a significant mean-level shift, but no significant variance reduction, suggesting authors became more lexically diverse on average without converging. The mean-level shifts observed in sentence length and hedge rate is consistent with known tendencies of RLHF-trained models, which have been shown to reduce linguistic variability in favour of perceived helpfulness and coherence (Sourati et al., 2026, p9-10), though the post-release period coincides with broader temporal factors that could independently drive these patterns, discussed further in the limitations section. If journalists are using LLMs as drafting tools rather than generating full text, one would expect exactly this pattern, with sentence length and hedging shifting toward LLM-characteristic norms while discourse marker usage and lexical diversity, being more deeply embedded in individual authorial voice, remain resistant to displacement. This aligns with Filimonovic et al. (2026, p.15), who find convergence in scientific writing to be partial rather than full.

Our second hypothesis predicted that author responses would be heterogeneous rather than uniform across the post-release period. This prediction is confirmed. Across all seven features, no linguistic feature exhibits a uniform directional shift, with authors varying considerably in both the direction and magnitude of their trajectories. This heterogeneity is most clearly illustrated by the contrasting patterns of sentence length and hedge rate. The negative intercept-slope correlation for hedge rate suggests that authors who hedged more before ChatGPT's release reduced their hedging more afterwards, while low-hedging authors increased theirs, a pattern consistent with genuine convergence toward a common epistemic register. The positive intercept-slope correlation for sentence length tells the opposite story, with authors who already wrote longer sentences tending to increase them further post-release, while shorter-sentence authors moved in the opposite direction. Interestingly, burstiness and MATTR-50 also display negative intercept-slope correlations, suggesting a convergence pattern in individual trajectories despite showing no significant mean-level shift in Model 2. This raises the possibility that convergence in these features is occurring at the individual level without yet manifesting in the corpus average, consistent with an early-stage adoption process. These bidirectional trajectories nuance Sourati et al.'s (2026, p.5) concern that LLMs risk flattening linguistic diversity, as our evidence suggests that if LLM influence is present, it is operating unevenly across authors and has not yet produced a flattening of linguistic diversity. Together, the heterogeneous trajectories observed here are consistent with Rogers' diffusion of innovation

framework, suggesting the corpus may be at an early and uneven stage of gradual adoption rather than uniform convergence.

4.2 Limitations

A key limitation concerns the observational design and absence of author-level predictors. In this study, the observed linguistic patterns may coincide with broader temporal factors such as shifts in editorial practice or platform pressures. Findings are therefore intentionally framed as temporal associations rather than causal claims. Incorporating author-level predictors such as direct LLM usage in future research would allow convergence to be tested among known adopters rather than inferred from linguistic signals alone, which could be achieved through journalist surveys to establish direct evidence of LLM usage.

A second limitation concerns the observation window. The post-release period spans approximately three years, which may be insufficient to capture the full effect of LLM influence if the mechanism operates through gradual habituation. Future research could adopt a longitudinal tracking approach to determine whether the early convergence signals observed here intensify, stabilise, or reverse as LLM adoption matures.

Finally, dictionary-based discourse marker extraction relies on context-independent matching, which does not distinguish between discourse-marking and non-discourse-marking uses of high-frequency function words such as "and", potentially introducing measurement noise. Future research could apply dependency-based methods to improve classification precision.

4.3 Implications and Future Research

This study demonstrates that multilevel modelling is a viable approach for detecting corpus-level linguistic change in journalism, providing outlet-specific quantitative evidence where existing work remains largely theoretical. The governance implications are significant. Even absent causal attribution, the temporal coincidence of linguistic change with LLM's public releases suggests that invisible linguistic drift may be occurring below the threshold of individual article detection. Current detection-based AI regulation is insufficient to capture gradual corpus-level shifts, and policymakers concerned with preserving journalistic diversity may need to consider variance-based monitoring frameworks that track linguistic change at the population level.

5. Conclusion

This study investigates whether linguistic convergence is detectable among Guardian Business journalists coinciding with the mass public releases of large language models. Using a multilevel modelling framework across 14,847 articles from 45 authors spanning 2019 to 2025, the results provide partial evidence of modest, feature-specific linguistic change in the post-ChatGPT period, most notably in hedge rate, which shows a consistent convergence pattern across both corpus-level and author-level analysis. However, random slope analysis reveals that individual author responses are heterogeneous and bidirectional, with convergence and divergence coexisting within the same corpus depending on the author and the feature. Consistent with the diffusion of innovation framework, these patterns suggest that if LLM influence is present, it is operating gradually and differentially across the corpus rather than converging uniformly. As LLM adoption deepens across professional writing contexts, variance-based monitoring of between-author linguistic change offers a promising framework for tracking the long-term effects of AI on authorial voice and editorial distinctiveness.

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7. Appendix

Appendix A

The discourse marker lists below are implemented using case-insensitive phrase matching. Multi-word expressions are treated as fixed phrases.

1. **Contrastive Discourse Markers:**

   but, alternatively, although, contrariwise, contrary to expectations, conversely, despite this, despite that, even so, however, in spite of this, in spite of that, in contrast to this, in contrast to that, instead of this, instead of that, nevertheless, nonetheless, notwithstanding, on the other hand, on the contrary, rather than this, rather than that, still, though, whereas, yet

2. **Elaborative Discourse Markers:**

   above all, after all, also, analogously, besides, by the same token, correspondingly, equally, for example, for instance, further, furthermore, in addition, in other words, in particular, likewise, more accurately, more importantly, more precisely, more to the point, moreover, on that basis, on top of it all, otherwise, rather, similarly

3. **inferential Discourse Markers:**

   all things considered, as a conclusion, as a consequence of this, as a consequence of that, as a result of this, as a result of that, because of this, because of that, consequently, for this reason, for that reason, hence, it follows that, accordingly, in this case, in that case, in any case, on this condition, on that condition, on these grounds, on those grounds, then, therefore, thus

Appendix B

The hedge expressions list below is implemented using case-insensitive phrase matching, with verbs matched in lemma form.

may, might, could, would, should, perhaps, possibly, probably, apparently, presumably, seemingly, arguably, generally, typically, usually, often, sometimes, about, around, approximately, roughly, nearly, almost, somewhat, suggest, indicate, appear, seem, tend, assume, believe, imply, possible, probable, likely, uncertain, unclear, approximate

Appendix C

Table A1: Summary Statistics for Linguistic Features

Variable	Mean	Standard Deviation	Median	Min	Max
Average Sentence Length	26.864	3.980	26.737	11.355	97.800
Burstiness	0.396	0.096	0.389	0.091	1.759
MATTR-50	0.821	0.019	0.822	0.592	0.890
Hedge Rate	12.698	6.536	11.990	0.000	45.699
Contrastive DM Rate	5.506	3.654	5.063	0.000	27.933
Elaborative DM Rate	27.907	8.712	27.211	0.000	139.059
inferential DM Rate	1.341	1.752	0.625	0.000	14.519

Table A1 presents summary statistics for all seven outcome variables in raw units, prior to transformation and standardisation. Statistics are computed at the article level across the full corpus of 14,847 articles.

Appendix D

Figure 3: Author-level Random Intercepts for Burstiness, Null Model

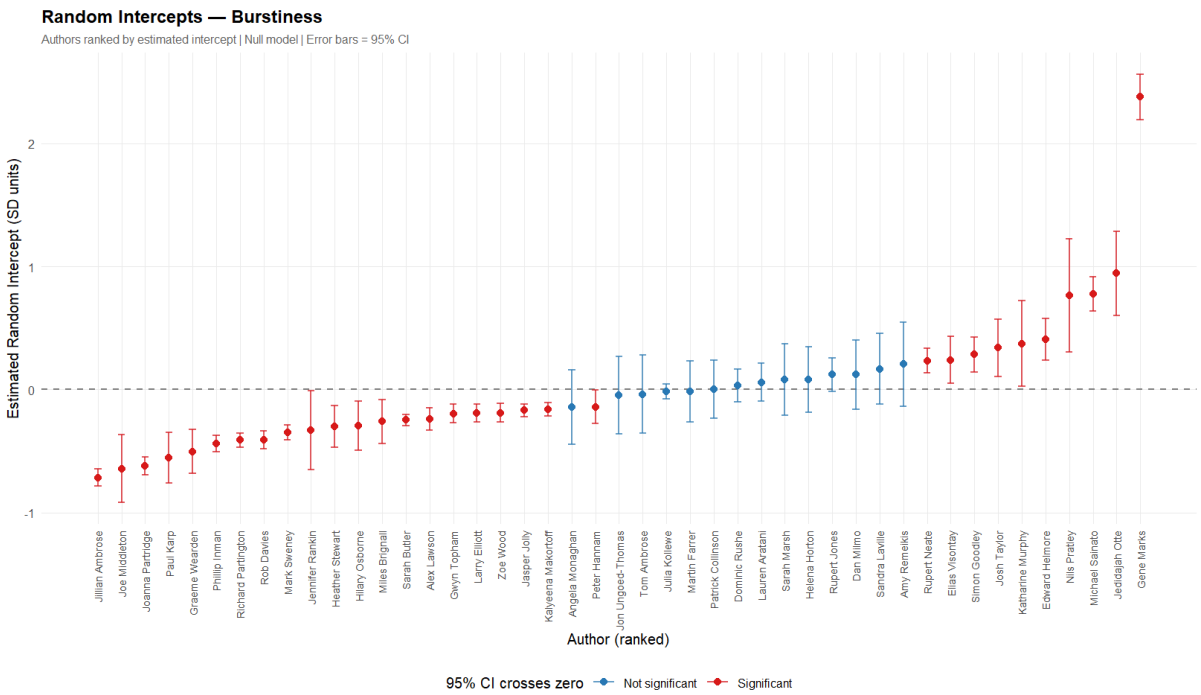


Figure 4: Author-level Random Intercepts for Contrastive DM Rate, Null Model

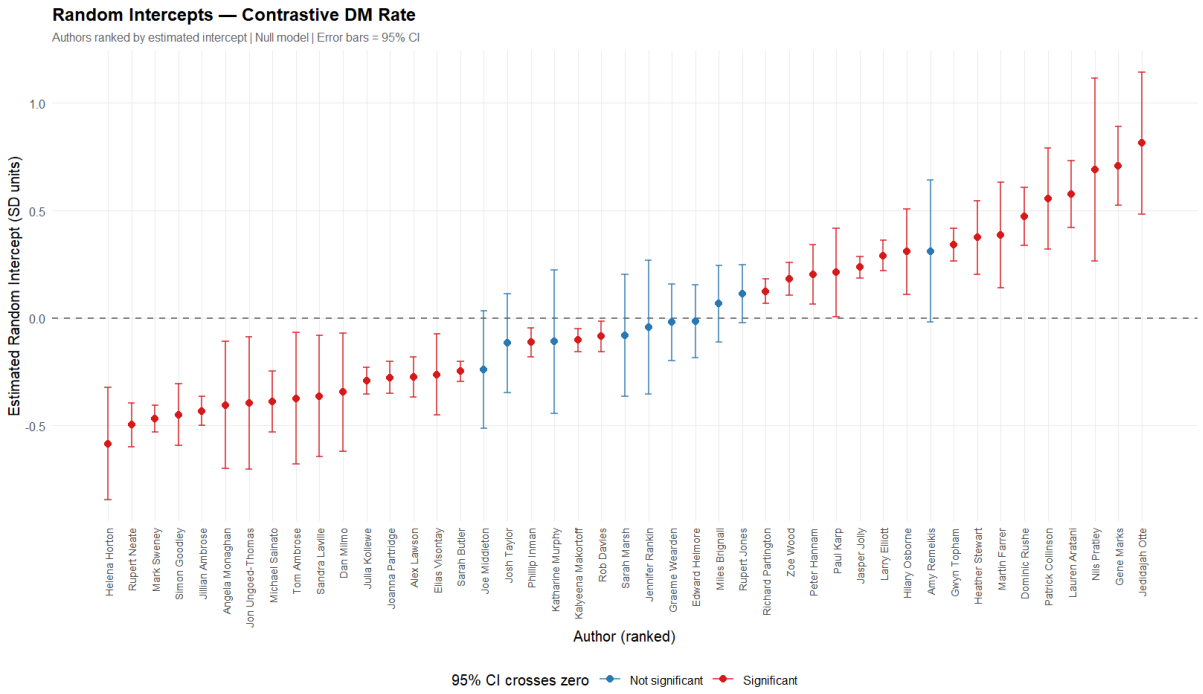


Figure 5: Author-level Random Intercepts for Elaborative DM Rate, Null Model

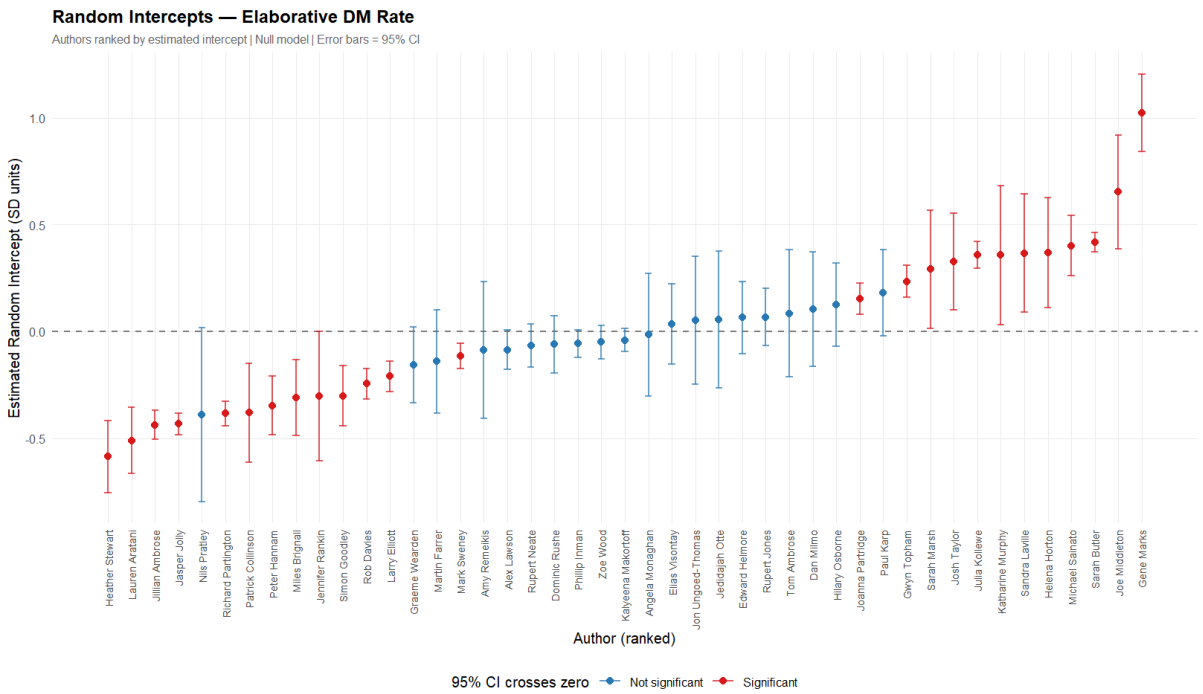


Figure 6: Author-level Random Intercepts for Inferential DM Rate, Null Model

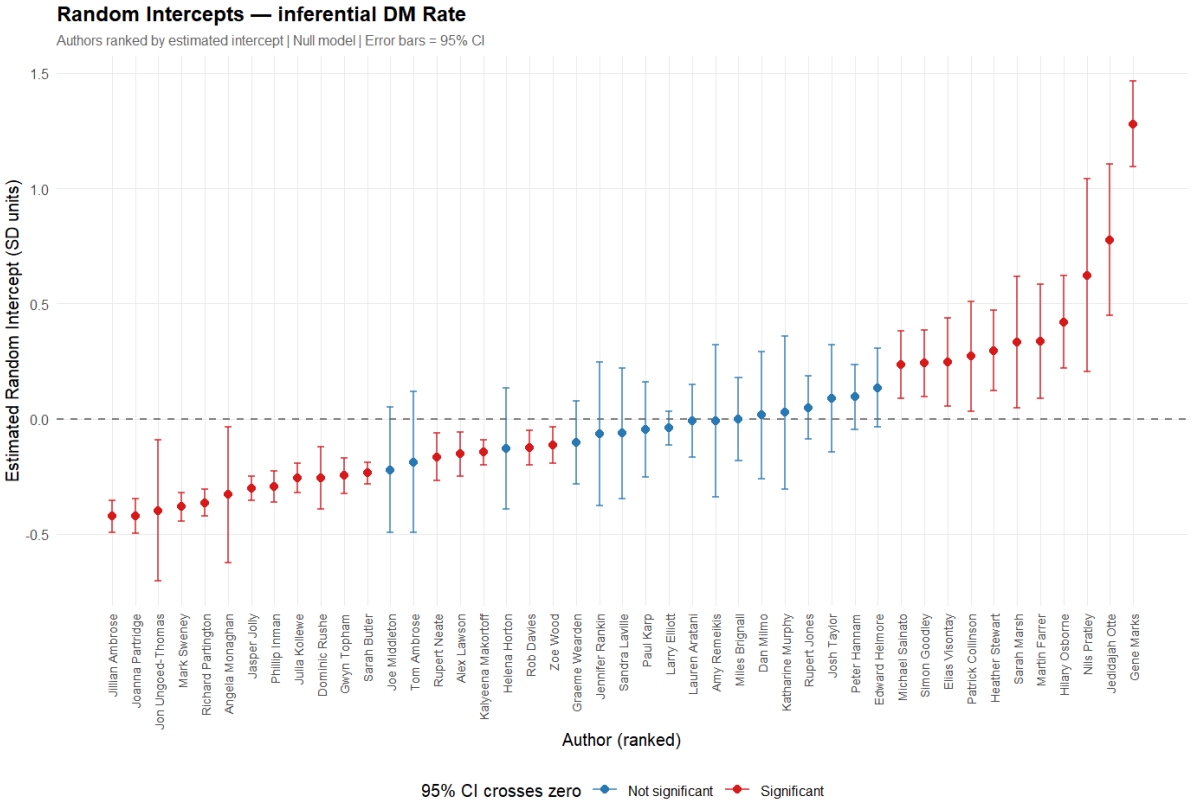


Figure 7: Author-level Random Intercepts for Hedge Rate, Null Model

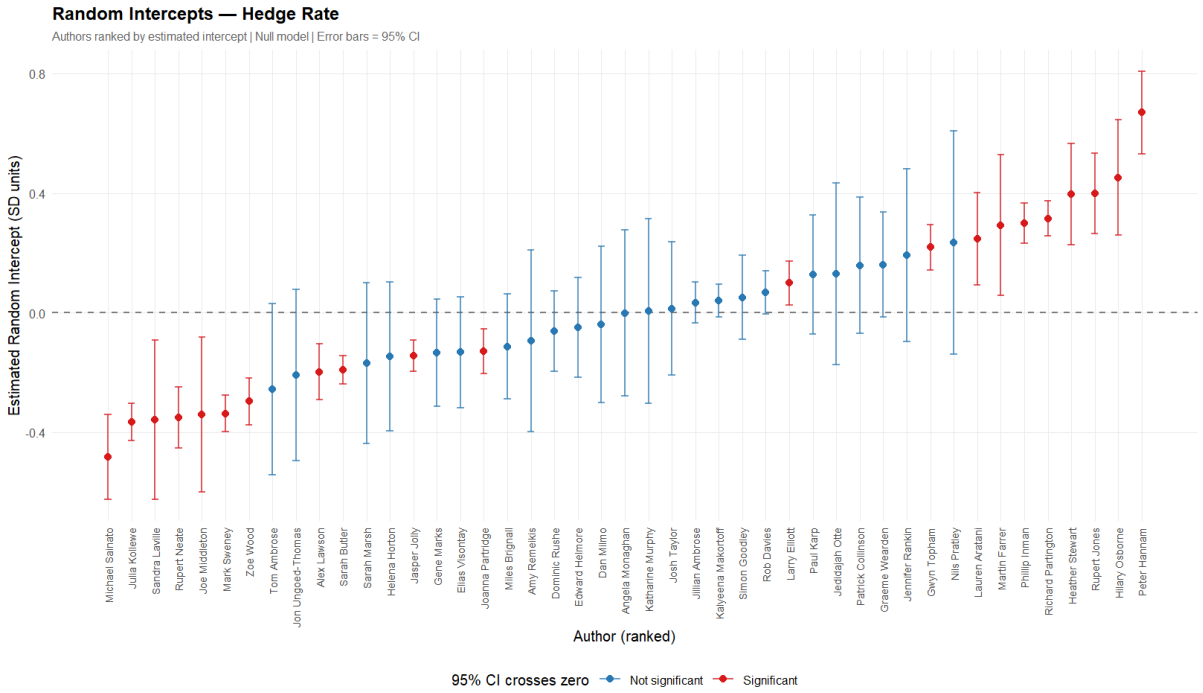
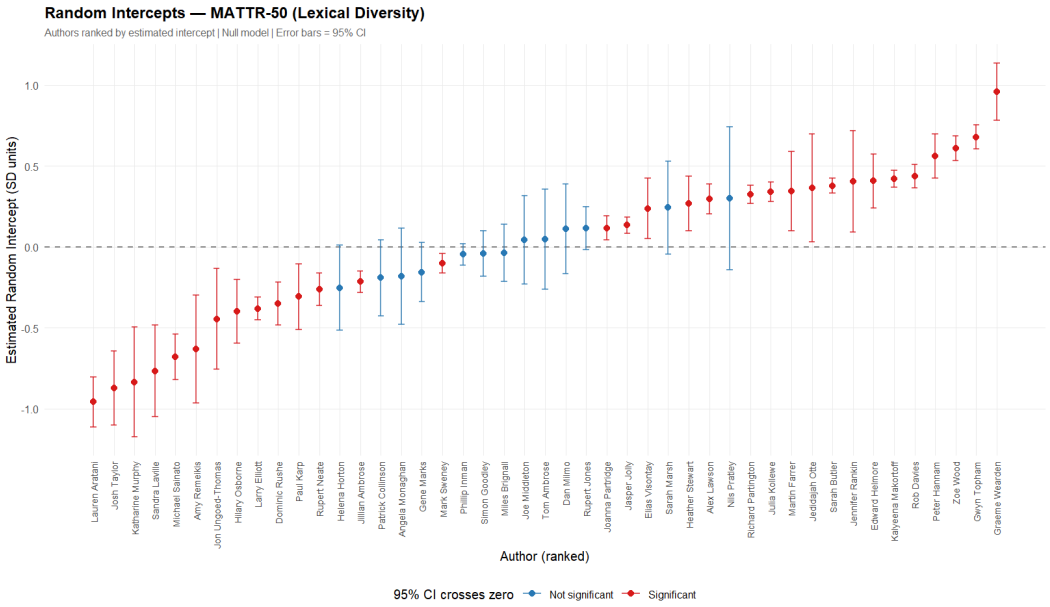


Figure 8: Author-level Random Intercepts for MATTR-50, Null Model



Appendix E

Table A2: Article Length Coefficient Estimates Model 2 Fixed Effects

Outcome	Estimated Coefficient $\hat{\beta}_2^W$	Standard Error	p-value	Estimated Coefficient $\hat{\beta}_2^C$	Standard Error	p-value
Average Sentence Length	-0.154	0.007	< 0.001	-0.129	0.049	0.012
Burstiness	0.272	0.008	< 0.001	0.158	0.044	< 0.001
Contrastive DM rate	0.163	0.008	< 0.001	0.084	0.035	0.022
Elaborative DM rate	0.004	0.008	0.606	0.011	0.034	0.750
Inferential DM rate	0.208	0.008	< 0.001	0.143	0.028	< 0.001
Hedge rate	0.069	0.008	< 0.001	0.047	0.027	0.084
MATTR-50	0.046	0.008	< 0.001	0.001	0.043	0.984

Note: $\hat{\beta}_2^W$ is the within-author effect of article length (each article's log word count demeaned by that author's mean). $\hat{\beta}_2^C$ is the between-author effect (each author's mean log word count). Both components are standardised.